

Title: Velum movement in speech and inter-speech pause intervals: A cineradiographic study of French and English speech

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34 **Velum movement during speech and inter-speech pause intervals: A**
35 **cineradiographic study of French and English speech**

36
37 **Abstract**

38 Velum behavior in speech production, particularly with nasal sounds, has been a
39 matter of significant interest to researchers revealing many different factors that
40 affect velum movement during speech events. Few studies, however, have
41 explored velum movement patterns during inter-speech pauses compared to
42 speech segments. To address this gap, we examined the velocity of velum
43 movement during the production of both nasal sounds and inter-utterance
44 pauses. We hypothesized that velum movement patterns differ between these
45 two contexts and that language background may modulate these patterns. We
46 analyzed velum movement in sentence-level speech of Québécois French and
47 English speakers from the Université Laval X-ray videofluorography database.
48 We measured the velopharyngeal opening (VPO) as the distance between the
49 velum's upper surface and the posterior pharyngeal wall. The change in VPO
50 over time served as a proxy for the velocity of velum movement during speech
51 segments and inter-speech pauses. We predicted faster velum movement during
52 speech pauses and also faster movement for English relative to French
53 speakers. Our results show that the velum behaves differently between speech
54 and pause events in terms of the velocity and duration of the movement. In
55 addition, velum behavior differs between languages, indicating language-specific
56 articulatory configurations for the velum.

57 **1 Introduction**

58 Although hidden from view, the function and characteristics of the velum as an active
59 speech articulator have gradually been brought to light. By the time Moll (1962)
60 investigated the effect of consonant context on vowels, it was already known that there
61 was greater velopharyngeal closure for high vowels than low vowels. Among other
62 aspects, previous studies have investigated how the size of the velopharyngeal opening
63 can be affected by segmental (Bell-Berti 1993, de Boer et al. 2023) and suprasegmental
64 features (Krakow 1993), whether the velum can act as an independent oral articulator
65 (Gick et al. 2013), how the velum's movement velocity can be related to other
66 articulators like the jaw or lips (Kollia et al. 1995), how the velum's movement velocity
67 can be affected by speech rate, sex, and velum size (Kuehn 1976), whether there is a
68 difference between the duration of the raising and lowering of the velum, or whether the
69 velocity of velum raising could be different from the velocity of velum lowering (Amelot
70 2004; Birkholz and Kleiner 2021).

While existing studies have explored velum movement velocity in speech, their focus was predominantly on the velum's behavior during speech segments. That is, to our knowledge, there has been no substantial attempt to study the behavior of the velum during inter-speech pauses. Knowledge of velum's behavior during speech pauses can offer valuable insights regarding whether the human body acts differently for speech-specific vs. speech-independent events. Therefore, this study seeks to investigate whether velum movement velocity differs between speech-related (e.g., nasalization of segments) and speech-independent (e.g., inter-speech pauses) tasks. Building on previous research (e.g., Islam et al. 2020), we hypothesize that velum movement velocity would be different for non-speech events compared to speech events.

The velum, also known as the *soft palate*, separates the oral and nasal cavities posteriorly and forms the anterior portion of the velopharyngeal port (VPP). During tidal breathing, the velum is typically lowered, leaving the VPP open. During speech, the velum is raised for oral sounds, restricting access to the nasal cavities by pressing against the (posterior) velopharyngeal wall, while for nasal sounds, the velum is lowered to allow sound to resonate in the nasal and sinus cavities. The velum also lowers for pauses and nasal breathing between utterances (Tierney and Bryson 2020; de Boer et al. 2023). The raising of the velum is due to the contraction of the U-shaped *levator veli palatini* muscle (LVP) which pulls the velum both upward and slightly backward and intrinsically contributes to the formation of a characteristic "bulge" on the posterior surface of the velum (Pigott 1969; Anderson et al. 2019). The *musculus uvulae* and the *superior pharyngeal constrictor* can also aid in closing the VPP (Perry 2011). The palatoglossus and palatopharyngeus muscles link the palate to the sides of the tongue and the stylopharyngeus respectively (Perry 2011). While the palatoglossus and palatopharyngeus are ideally positioned to pull the velum down (Fritzell 1969), both are attached to soft tissue, and EMG studies have not shown consistent activation prior to nasal sound production (e.g., Bell-Berti 1976; Benguerel et al. 1977b). The lowering of the velum is thought to result from suppression/deactivation of the LVP and the force of gravity, perhaps aided by elastic tissues within the anterior faucal arches and *palatoglossus* (Bell-Berti 1993; Kuehn and Azzam 1978).

In the analysis of velum movement, some authors have focused on velum raising and lowering (e.g., Benguerel et al. 1977a), others on VPP opening and closing (e.g., de Boer et al. 2013) and others on both (e.g., Moll and Daniloff 1971). While velar height and the area of the VPP are highly correlated (Ushijima & Sawashima 1972 - as cited by Bell-Berti 1980) it's not a perfect one-to-one relationship. Once closure is made, the area of the VPP is zero, but with tighter closure, the height of the superior surface of the posterior portion of the velum can continue to increase (Moll & Daniloff 1971) forming the velar eminence (i.e., velar "knee") (Perry 2011) with increased LVP activation (Anderson et al. 2019).

A number of factors have been found to affect the velocity of velum movement; this includes the direction of movement (which Kuehn (1976) refers to as “phonetic context”), travel distance, speech rate, and sex (Amelot 2004; Benguerel et al. 1977a; Birkholz and Kleiner 2021; Kollia et al. 1995; Kuehn 1976; Martins et al. 2012; Oliveira et al. 2009; Teixeira and Vaz 2001), among others. For direction of movement, raising and lowering the velum (closing and opening the VPP), some studies have found no difference in velocity for some of their speakers (Birkholz and Kleiner 2021; Kuehn 1976), while other studies found faster velocities for raising/closing movements than lowering/opening movements (e.g., Benguerel et al. 1977a; Birkholz and Kleiner 2021; Kuehn 1976). Studies that measured the duration of movement instead of velocity also reported conflicting results with either the duration of raising and lowering the velum being the same (Amelot 2004) or the duration of velum raising was shorter than velum lowering (Martins et al. 2012; Oliveira et al. 2009; Teixeira and Vaz 2001). Studies have shown a positive association between velum movement velocity and the travel distance by the velum during an event. Kuehn (1976) and Kollia et al. (1995) reported that the velum moved faster when the distance or displacement was greater.

For the effect of speech rate, an increased speech rate has been found to be positively correlated with velum movement velocity; that is, an increased speech rate involved faster movement of the velum (Kuehn 1976); this phenomenon also involved a shorter travel distance, which Kuehn attributed to articulatory “undershoot” (pg. 317). As for sex, Kuehn (1976) found that the male speaker moved the velum faster than the female speaker, while Birkholz and Kleiner (2021) inferred that for the two male speakers, the velum moved faster for raising than lowering (with lateral wall movement serving as a proxy for velar movement), but not for the three female speakers. Similar differences between male and female speakers have been reported in Amelot (2004) as well. Overall, the velocity of velum movement has been tied to multiple language-independent factors, as described above.

Another source of velum movement velocity variation can be the language being spoken. The studies cited above were among speakers of English (Kollia et al. 1995; Kuehn 1976), French (Amelot 2004; Benguerel et al. 1977a), German (Birkholz and Kleiner 2021) and European Portuguese (Martins et al. 2012; Oliveira et al. 2009; Teixeira and Vaz 2001). The inconsistencies in the findings of these studies, in terms of whether or not the velocity of velum raising (closing) is faster than velum lowering (opening), could be attributed to their small sample sizes, often of only one or two speakers, but may also be due to the language spoken. Comparisons between these four languages are not possible due to differences in how the velocity of velum movement was calculated in the respective studies. A particular aspect of languages that may affect velum movement velocity is phonemic vowel nasalization. French and Portuguese have both coarticulatory nasalization of oral vowels and phonemically nasal

vowels, while English and German have only coarticulatory vowel nasalization.

The size of the velopharyngeal opening (VPO) in nasal sound production can also be affected by language differences. For example, Islam et al. (2020) reported that Canadian English nasal segments have a larger relative VPO compared to Quebecois French nasals. Assuming a relatively constant speech rate, a larger VPO would be predicted to correlate with a faster movement of the velum toward its target position, since evidence suggests the larger the VPO, the faster the velum moves (Kuehn 1976; Kollia et al. 1995). Based on the knowledge of the difference between English and French nasals in terms of VPO size, we also examined whether language difference can have an effect on the velocity of velum movement in speech as well as in pause intervals.

A further possible source of velum movement variation is the dialects of the languages, such as between Quebecois and European French. Belgian, Swiss and Quebecois French have four nasal vowels /*ɑ̃*/, /*ɛ̃*/, /*ɔ̃*/ and /*œ̃*/ (albeit in Quebecois French /*ɑ̃*/ is often pronounced [ã] and /*ɛ̃*/ as [ẽ]) while in Standard Parisian French there are only three as /*ɛ̃*/ and /*œ̃*/ merged (Bigot & Papen, 2013). But there are also differences in coarticulatory nasalization, Quebecois French has greater VPO for oral vowels preceding nasal consonants (de Boer et al. 2023), while for Belgian and Standard French carryover nasalization is more prominent (Basset et al., 2001; Delvaux et al., 2008). As for English, Cunha et al. (2024) recently found that anticipatory vowel nasalization was more extensive in speakers of American English than standard Southern British English. However there was no differences in the size, peak lowering velocity, or articulatory duration of the velum movement between the two groups (Cunha et al. 2024).

As briefly stated above, another aspect of articulatory movements we investigated was the movement of the velum during speech-related vs. physiological movements. We defined the speech-related movement as the velum activity during a speech interval.; For example, the opening of the VPP (by means of lowering the velum) to produce nasality in a nasal sound like [ɔ̃], [m], or [n], etc. was categorized as speech-related velum movement, as was the closing of the VPP for a following oral sound. On the other hand, the physiological movement category included the lowering of the velum (and its subsequent raising) during the pause interval between two utterances. For example, a lowering followed by a raising of the velum for the purpose of breathing in between two chunks of speech would qualify as a physiological movement. The motivation behind this method of categorization was to explore whether the velum behaves differently between the speech domain and the physiological domain, and, if yes, how.

An archival X-ray study was thus conducted to test the hypothesis that the velum behaves differently in speech-related vs. speech-independent events since speech

189 requires specific gestural targets, unlike non-speech movements; specifically, we
 190 predict that the velum will move faster for movements during speech segments
 191 compared to inter-utterance pauses. This archival data set also enables us to test the
 192 prediction that velum movement should differ between French and English, as
 193 described above.

194 **2 Method**

195 **2.1 Subjects**

196 The speech samples used in this study were from 9 speakers (4 females and 5 males)
 197 of Quebecois French and 4 speakers (3F, 1M) of Canadian English obtained from the
 198 Université Laval X-ray videofluorography database (Munhall et al. 1995). For French,
 199 the total number of X-ray video files was 17; for English, there were 7. Table 1 provides
 200 more information (based on Munhall et al. 1995) on how many files came from each
 201 speaker. The age range for French speakers was 19-30 years; for English speakers, the
 202 range was 22-38 years. All French speakers were from Quebec; three of the English
 203 speakers were from the province of British Columbia and one was from New Brunswick.
 204 While the speakers from British Columbia can be assumed to be speaking Canadian
 205 English, which is spoken in central and western Canada, however, the speaker from
 206 New Brunswick may have been speaking Atlantic English. For the sake of simplicity we
 207 will be treating the four as speakers of Canadian English.

208 Table 1: Demographic information of the subjects (based on Munhall et al. 1995)

Language	Speaker ID	Film ID	sex	Number of films
French	Fr1	24-25	F	2
	Fr2	26-27	F	2
	Fr3	29	M	1
	Fr4	30-31	F	2
	Fr5	32-33	M	2
	Fr6	43	M	1
	Fr7	44-45	M	2
	Fr8	48-50	M	3
	Fr9	55-56	F	2
	En1	77	M	1
	En2	73-74	F	2

English	En3	78-79	F	2
	En4	80-81	F	2

209

210 **2.2 Data**

211 The Université Laval X-ray videofluorography database, the source of the data in this
 212 study, was originally developed by Dr. Claude Rochette in 1974 in the form of X-ray
 213 films (50 frames per second); the database was later digitized by Munhall et al. (1995)
 214 at a rate of 30 frames per second. In the process of digitization, the accompanying
 215 audio was stretched to synchronize with the video by adding audio samples, which
 216 maintained the pitch of the original recording. More specific technical details of the
 217 digitized database can be found in Munhall et al. (1995).

218 The database comprises speech recordings from native English and French speakers,
 219 each producing short sentences—typically six syllables long—in their first language.
 220 These sentences were carefully constructed to include a wide variety of phoneme
 221 combinations. As each sentence was read individually, natural pauses occurred
 222 between consecutive chunks of speech. The dataset includes all types of phonologically
 223 nasal segments—both consonantal and vocalic—distributed across various
 224 phonological positions, including word-initial, medial, and final. For pause tokens, the
 225 onset was annotated at the end of one sentence, and the offset at the beginning of the
 226 next. Below are a few examples of the sentences used as stimuli.

- 227 • French: "Ce talus ombrage le chemin."
- 228 • French: "Lit un bouquin inutile."
- 229 • French: "Le néant s'est obscurci."
- 230 • English: "The crowd was tumultuous."
- 231 • English: "Gargoyles are interesting."
- 232 • English: "A partition was built."

233 We used the accompanying audio files in the database to manually transcribe
 234 sentences and phonemes into Praat TextGrids (Boersma & Weenink, 2025). However,
 235 no acoustic analyses were conducted using the audio in this study.

236 For our data, we measured the size of the opening in the velopharyngeal port (VPP)
 237 over time. That is, we measured the distance between the posterior pharyngeal wall and
 238 the upper surface of the velum every 33.3 milliseconds (to match the frame rates used
 239 in the digitized database) for the whole length of the video for each speaker; we called
 240 this measure Velopharyngeal Opening (VPO). To measure the VPO, we used ImageJ
 241 (Rasband 2018; Schneider, Rasband and Eliceiri 2012) to fit a profile line along the path

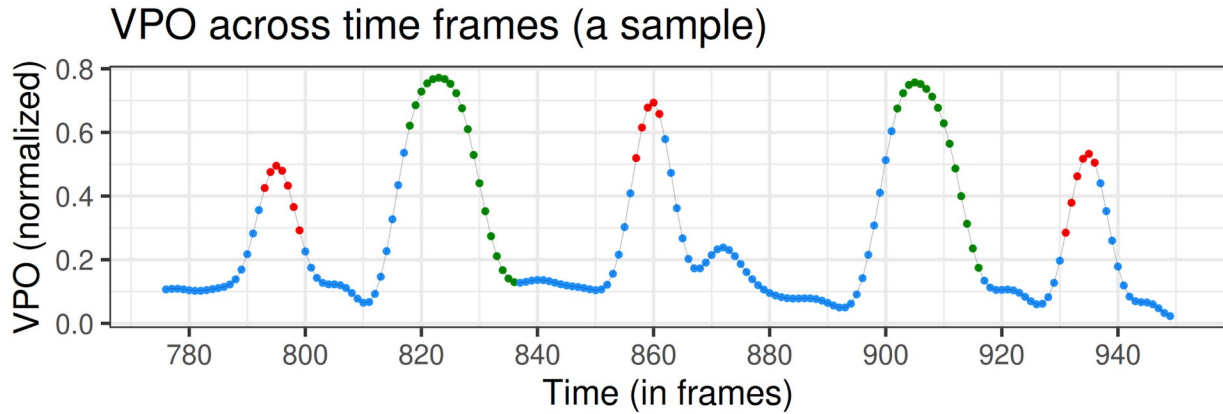
242 of velum movement (see Figure 1) between the pharyngeal wall and the upper surface
 243 of the velum. This profile line corresponded to the middle of the LVP muscle which pulls
 244 the velum towards the PPW, and so, was diagonal (between 30 and 60°) to the vertical
 245 plane of the vocal tract. The start of the profile line was placed on the PPW at the height
 246 where the velum first made contact with the PPW during closure. The end of the line
 247 was placed on the superior surface of the velum where that contact was first made.



248
 249 Figure 1: The path of velum movement (video file: laval80, frame taken at 44 sec
 250 through the video file.)

251 With the path of the velum traced, we measured the number of black pixels along the
 252 velum path. The measurements were taken semi-automatically; for each video file, line
 253 along the path of velum movement was determined manually and then the process of
 254 counting the pixels for all the frames from a video file was performed automatically with
 255 ImageJ. The process did not involve any additional filter since the images were already
 256 black-and-white. A larger number would indicate a wider opening in the velopharyngeal
 257 port (i.e., greater VPO), as expected in a nasal speech segment (e.g., [n]) while a
 258 smaller opening would indicate a smaller VPO, as it would be expected in an oral
 259 speech segment (e.g., [a]). The absolute VPO values (number of black pixels) were
 260 normalized between 0 and 1 within each video file so that a VPO of 1 represents the
 261 speaker's widest opening in the velopharyngeal port while a value of 0 represents no
 262 opening. All normalization procedures were performed within individual speakers,
 263 across all of their utterances (speech and non-speech intervals). (More specific details
 264 of this level of processing can be found in de Boer et al. 2023) Figure 2 provides an
 265 example of the VPO values over time from a French speaker. In the figure, the x-axis
 266 refers to the time (with frame number as a proxy) and the y-axis refers to the

267 corresponding VPO. The three colors represent three types of segments/events: Oral
 268 vowels and consonants (O), Nasal vowels and consonants (N), and Pause intervals (P).
 269 As the figure shows, the VPO is greater during both nasal and Pause events while the
 270 VPO remains considerably smaller during oral segments.



271
 272 Figure 2: VPO (normalized) as a function of time (as frame number). The peaks indicate
 273 wider openings in the VPP while values around the lower region indicate relatively
 274 closed VPP. Colors: blue = oral sounds, red = nasal sounds, green = inter-speech
 275 pause.

276 2.2.1 Velocity of velum movement

277 Since we were more interested in the velocity of the velum movement than in the
 278 distribution of VPO values across different events, we used the normalized VPO values
 279 and the corresponding frame numbers to calculate the slopes during the rising and the
 280 falling phase of an opening or closing event during the velum's movement. We use the
 281 term "velum cycle" to refer to a full cycle of velum movement consisting of an opening
 282 and a closing phase; for example, in Figure 2, the pattern between frames 850 and 867
 283 is a velum cycle since it has an opening phase (the velum starts from a closed position
 284 to approach an open position) following by a closing phase (the velum comes from an
 285 open position back to a closed position), thereby completing a full cycle of movement.
 286 We also use the term "peak" to refer to the point that has the highest VPO during the
 287 length of a velum cycle and the term "valley" to refer to the lowest points on the two
 288 sides of a peak.

289 Next, slopes were calculated independently for the two types of events during a velum
 290 cycle: 1) the opening phase and 2) the closing phase. For example, in Figure 2, one
 291 instance of the opening phase can be found approximately between frames 850 and
 292 860. A simple linear regression model was fit with the data points during the opening
 293 phase (the "rising" phase in Figure 2) and the slope value was extracted from the
 294 model; this slope value was used as a proxy for the velocity of the VPP opening.

295 Similarly, another simple linear regression model was fit with the data points during the
 296 closing phase (the “falling” phase in Figure 2) and the slope value was used as a proxy
 297 for the velocity of closure of VPP. Since we were interested in the velum activity during
 298 nasal speech segments and between-utterance pause intervals, we excluded the
 299 occasional smaller velum cycles that had an oral speech segment (oral vowels or
 300 consonants) at its peak. More details of the inclusion and exclusion criteria are provided
 301 in the sections below.

302 **2.2.2 Duration**

303 For an opening or closing phase of a velum cycle, the duration value was calculated by
 304 counting the number of frames between the valley point and the peak. For example, in
 305 Figure 2 there is an opening phase between frames 850 and 860; the data point with
 306 the lowest VPO (the y-axis value) was defined as the first data point while the one with
 307 the highest VPO was defined as the last data point. In this particular case, the duration
 308 was about 10 frames (the number of frames between the valley and the peak).
 309

310 It is to be noted here that the measurements we obtained or calculated are more directly
 311 related to the size of the opening in the velopharyngeal port, rather than the raising or
 312 lowering of the velum itself. The size of the VPO should indeed be correlated with how
 313 high the velum is positioned (raised or lowered), and, even though we didn't track the
 314 movement of the velum directly, we are using the VPO measure as a proxy of the
 315 velum's raising and lowering activity in this paper.
 316

317 **2.2.3 Travel distance by the velum**

318 We also measured the maximal distance the velum travels during the oral, nasal, and
 319 pause events and we used the peak VPO for this purpose. More specifically, travel
 320 distance referred to the distance starting at a closed state of the VPO and ending at the
 321 velum position when the VPO was the largest during an open-close event. Thus, travel
 322 distance is a measure of how far the velum travels during an open-close event from the
 323 speaker-specific baseline (completely closed-state of the VPP).

324 **2.2.4 Inclusion and exclusion criteria**

325 Since we were only interested in the velum's activity during Nasal and Pause events,
 326 we excluded the data points that did not meet our criteria. Following are some more
 327 specific details of the inclusion/exclusion criteria:

- 328 1. A velum cycle was included only when there was a NASAL segment (shown as
 329 “N” in the figure) OR a “Pause segment” (represented by “P” in the figure)
 330 appearing around its peak. The nasal segments included /m, n, ŋ/ for English and
 331 /ɑ̃, ɛ̃, ɲ, m, n, ɔ̃, œ̃/ for French.

- a. The objective was to include only those VPP opening gestures where the targets were either a nasal sound or a pause (e.g., for breathing)
 - b. Any peak that had other speech sounds was excluded.
 - c. Also excluded were the velum cycles where a sequence of two Ns or a sequence of N and P (or vice-versa) appeared at or near the peak to avoid any possible confounding effects.
2. Velum cycles with very small fluctuations in rising and falling were also excluded; this was done by choosing an arbitrary value for how low the VPO of the peak should be; any peak that appeared below a VPO of 0.15 was automatically excluded.

2.3 Statistical analysis

2.3.1 Variables

Our final dataset included the following variables and levels of measurement:

- `speaker`: speaker ID
- `sex`: female vs. male (categorical)
- `language`: French vs. English (categorical)
- `interval_type`: Nasal vs. Pause (categorical). The value “Nasal” was recorded when the peak VPO during a velum cycle represented the opening in a nasal sound; the value “Pause” was recorded when the peak VPO represented a Pause/pause interval between two chunks of utterance.
- `phase`: opening vs. closing (categorical). Here, “opening” refers to the interval when the VPP opens (i.e., velum lowers); “closing” refers to the interval when the VPP closes (i.e., velum raises).
- `peak_distance`: the peak VPO value (the highest point) in an opening or closing phase (numeric)
- `duration`: the time taken (in frame counts) for an opening or closing phase (numeric)
- `velocity`: the slope obtained from an opening or closing phase (numeric)

2.3.2 Token counts per language

Table 2 provides a count of the total number of tokens, grouped by language and interval type, that were analyzed.

Table 2: Number of tokens per language and peak segments

Language	Peak segment	Count
English	Nasal	172
English	Pause	279
French	Nasal	576
French	Pause	696

366

367

368 2.3.3 Statistical analysis

369 Since there were multiple observations from each subject, we analyzed the data with
370 linear mixed-effects models which can take care of the random effects introduced by
371 speaker differences. We used R (R Core Team 2024) and *lmerTest* (Kuznetsova, et al.
372 2017) to perform a linear mixed-effects analysis of the relationship between velocity and
373 the relevant variables in the dataset: distance, duration_in_frames_log, language, sex,
374 opening_vs_closing_phase, and interval type. As fixed effects, we entered all these
375 variables (without any interaction terms) into the model. For random effects, the model
376 had intercepts for speakers. To make the weights of the numeric variables comparable,
377 the numeric predictor variables were scaled by two standard deviations using the
378 `r2sd()` function from the R package `stevenmisc` (Miller 2022); this converted the
379 variable to have a mean of 0 and a standard deviation of 0.5. The categorical predictors
380 were converted to dummy variables re-coding them into 0 and 1. This process makes
381 the resulting coefficients from numeric predictors directly comparable with the binary
382 predictors (Gelman 2008). Model diagnostics were performed via visual inspections of
383 the residual plots which showed that no obvious deviations from homoscedasticity or
384 normality were present. P-values were obtained via *lmerTest* package which “provides
385 p-values in type I, II or III ANOVA and summary tables for linear mixed models via
386 Satterthwaite’s degrees of freedom method” (Kuznetsova, et al. 2017, pg. 1).

387 3 Results

388 We report the results of this study in the following sub-sections below:

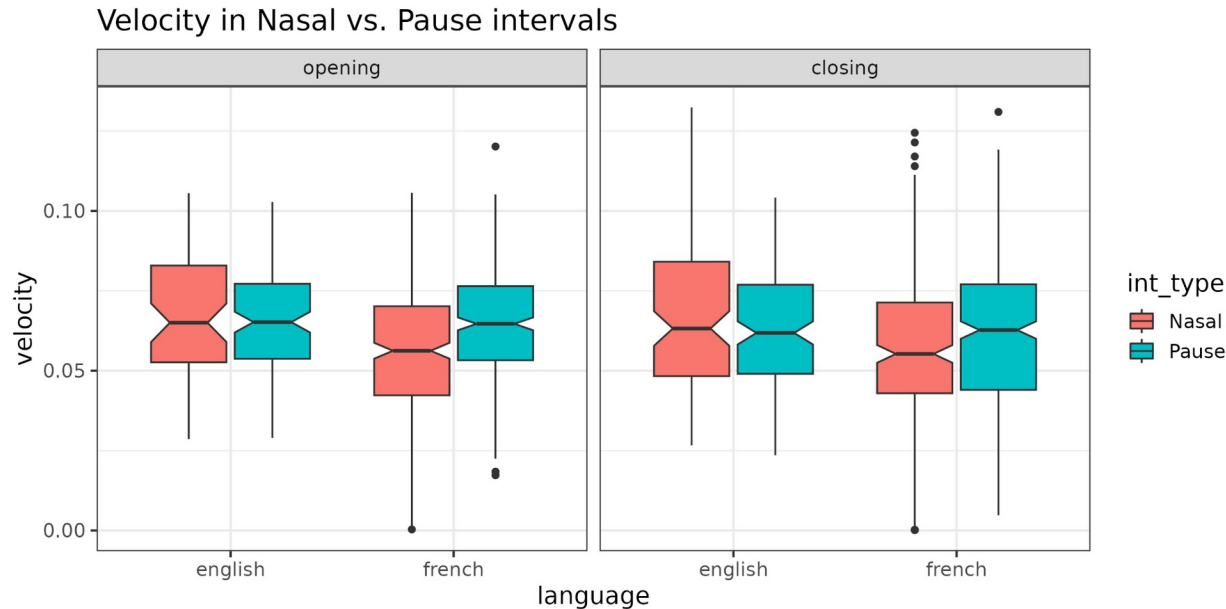
- 389 ● Velum movement velocity differences by speech (nasal segment) vs. non-speech
390 (pause) events
- 391 ● Relationship between velum movement velocity and travel distance
- 392 ● Relationship between velum movement velocity and duration of the movement

393 For each of these subsections, we first report the general summaries of the findings in
394 the form of visuals and summary statistics. We then present statistical assessments of
395 the effect of the related predictor variables on the dependent variable (velum movement
396 velocity) via linear mixed-effects models.

397 **3.1 Velocity**

398 **3.1.1 velum movement velocity in pause vs. speech**

399 Figure 3 presents the distributions of the velocity values in Nasal and Pause intervals in
400 the dataset. The y-axis represents the velocity of the velum movement whereas the x-
401 axis shows the language categories; the Nasal intervals are shown in coral-pink and the
402 Pause intervals in turquoise-blue. Data for the opening vs. closing phases of the VPP
403 are shown in different panels. As the figure indicates, the velum appears to move faster
404 in Pause intervals than in Nasal segments in French; this is true for both the opening
405 and closing phases. This difference seems more pronounced in the opening phase.
406 However, Pause intervals in English do not appear to differ from Nasal intervals in terms
407 of velocity in either the opening or closing phase. Summary statistics for velocity are
408 provided in Table 3 below.
409



410
411 Figure 3: Boxplot of velum movement velocity in Nasal vs. Pause intervals for opening
412 and closing in English and French.
413

414 Table 3: Mean velar velocity by language, phase, and interval type

language	phase	segment	mean	sd
English	closing	nasal	0.067	0.024
English	closing	pause	0.063	0.019
English	opening	nasal	0.066	0.020
English	opening	pause	0.064	0.016

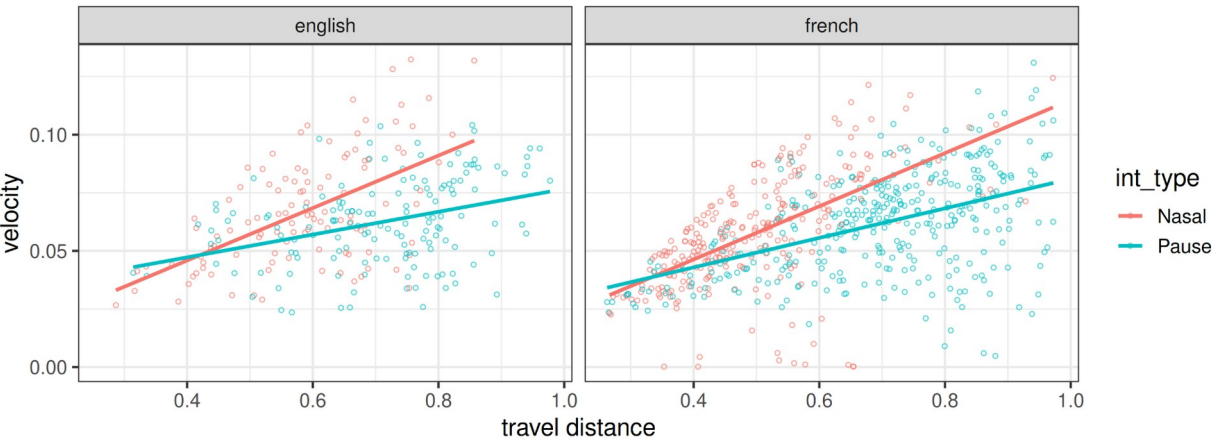
French	closing	nasal	0.057	0.024
French	closing	pause	0.061	0.022
French	opening	nasal	0.056	0.021
French	opening	pause	0.065	0.017

415

416 **3.1.2 velum movement velocity and travel distance**

417 This section investigates the relationship between the velocity of velum movement and
418 the distance the velum travels during the opening or closing phase. Figures 4 and 5
419 present this relationship for the opening and closing phases separately. In both figures,
420 the x-axis represents the distance the velum traveled during an opening or closing
421 phase (where a value of 1.0 refers to the maximum displacement of the velum from its
422 closed position). The y-axis represents the slope values obtained from simple linear
423 regression models (as described in Section 2.2 Methods section above). The straight
424 lines indicate the least-squares regression line for each category of data.

Velocity and travel distance of velum during closing phase



425

426 Figure 4: Scatterplot of velum movement velocity and travel distance in the closing
427 phase in English and French for nasal and pause intervals

428

429

430

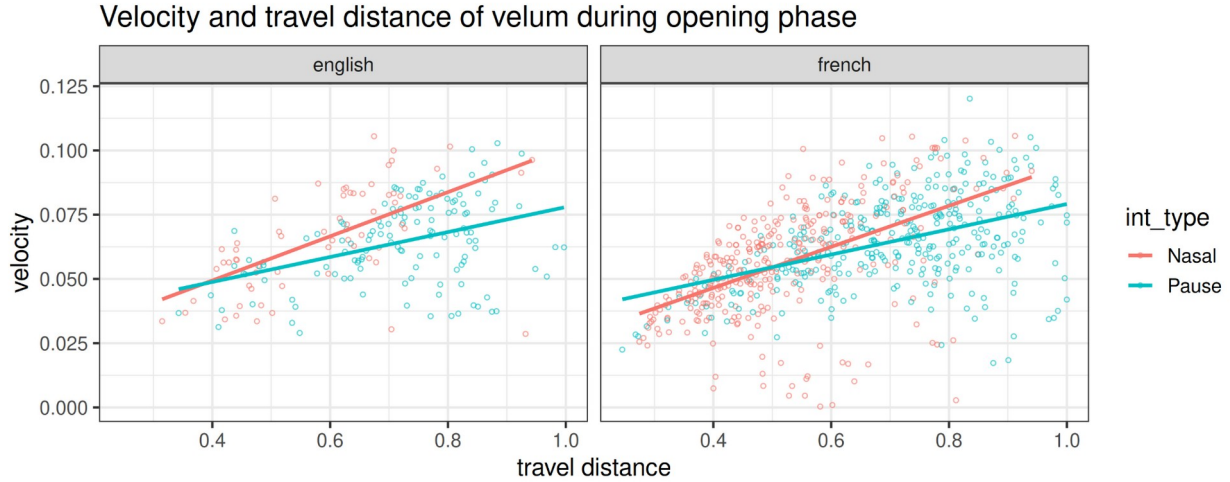


Figure 5: Scatterplot of velum movement velocity and travel distance in the opening phase in English and French for nasal and pause intervals

As all the panels in both figures indicate, there appears to be a positive relationship between the velocity of the velum's movement and the distance the velum travels; the greater distance the velum has to travel, the faster the velum moves. Also, the slope of the regression line is steeper in the Nasal intervals than in the Pause intervals, and the trend is seen for both opening and closing phases across both languages. Pearson's r values obtained from correlation tests confirm this trend; Table 4 presents the results of the correlation analyses within subsets of data. As seen in the table, Pearson's r was positive for all the combinations and the correlation coefficient was statistically highly significant in every case. The correlation was mostly of moderate strength (between .3 and .7), and the r was greater for the nasal category than for the Pause category across all the combinations.

Table 4: Correlation analysis between velocity and travel distance for velar opening and closing movement in English and French for nasal and pause intervals.

phase	language	segment	t	df	r	p_val
closing	English	nasal	7.04	107	0.56	<.001
closing	English	pause	4.59	146	0.36	<.001
closing	French	nasal	12.09	259	0.6	<.001
closing	French	pause	10.35	361	0.48	<.001
opening	English	nasal	5.86	61	0.6	<.001
opening	English	pause	4.99	129	0.4	<.001
opening	French	nasal	10.74	313	0.52	<.001

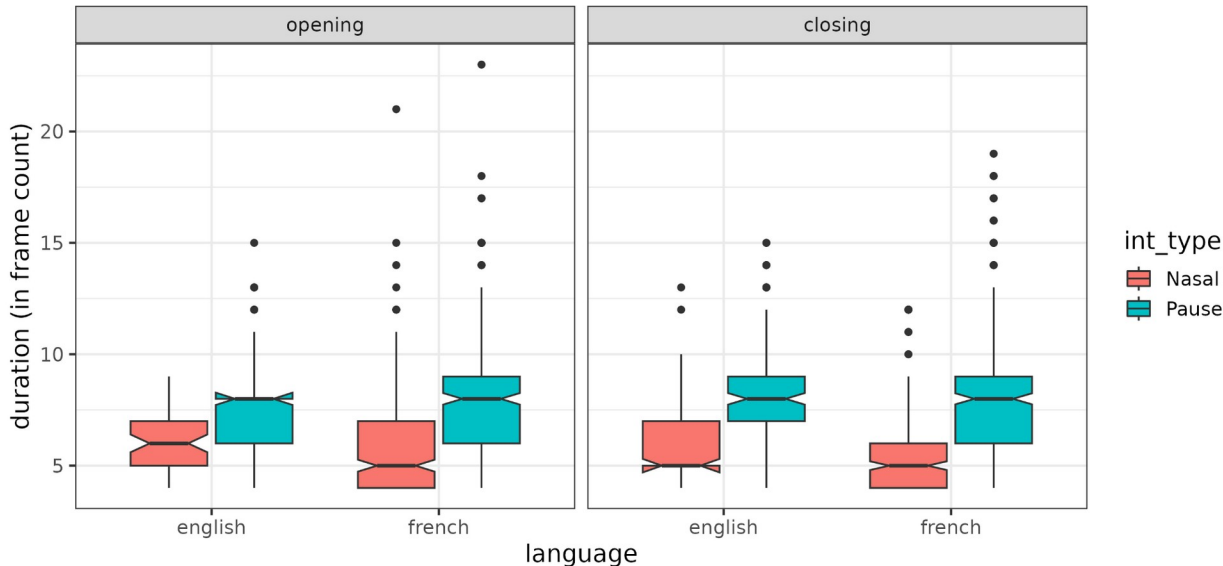
opening	French	pause	9.19	331	0.45	<.001
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449 **3.2 Duration**

450 **3.2.1 Duration in pause vs. speech**

451 Figure 6 displays the distribution of duration values for Nasal and Pause intervals in the
452 dataset. The y-axis represents the duration in frame counts, while the x-axis indicates
453 the language categories. Separate panels present data for the opening and closing
454 phases of the VPO. As indicated in the figures, Pause intervals have a consistently
455 higher number of frames than nasal intervals; this is true for both opening and closing
456 phases in both English and French languages. This suggests that the velum spent more
457 time traveling between the open and closed states during Pause intervals than Nasal
458 intervals. Summary statistics for duration are provided in Table 5 below.

Duration of movement in Nasal vs. Pause intervals



459
460 Figure 6: Box plot of duration of velar movement for Nasal vs. Pause intervals

461
462 Table 5: Summary statistics for duration (in frame count) of velar opening and closing
463 movement in English and French for nasal and pause intervals.

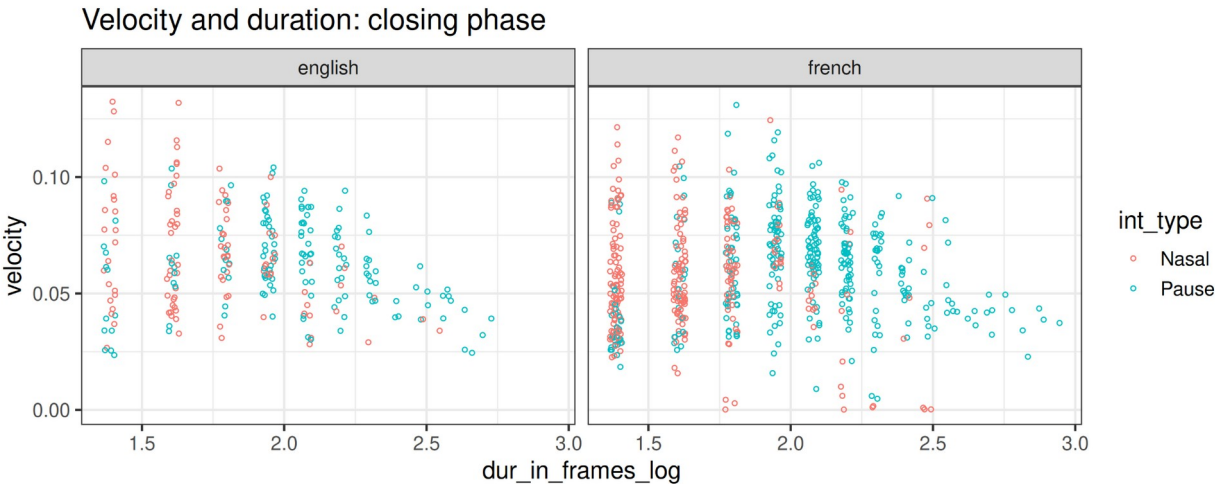
language	phase	interval	mean	sd
English	closing	nasal	5.84	1.71
English	closing	pause	7.92	2.47
English	opening	nasal	5.75	1.2
English	opening	pause	7.52	2.03

French	closing	nasal	5.46	1.75
French	closing	pause	8.01	2.61
French	opening	nasal	5.86	2.12
French	opening	pause	8.04	2.58

465

466 **3.2.2 Relationship between velocity and duration**

467 Similar to our investigation of the relationship between velar velocity and travel distance,
468 we also investigated the relationship between velocity and duration (since the velocity of
469 an object’s movement can be affected by the time it has to reach a certain point).
470 Figures 7 and 8 present the velar velocity values against the duration values, both for
471 the opening and closing phases. The x-axis refers to the duration (in frames) of an
472 opening or closing phase in logged form; the values were logged (natural log) to reduce
473 the skewness observed in the distributions of the data. The y-axis represents the
474 velocity of the movement. To avoid overplotting in the plot (primarily because of the
475 discrete nature of the frame counts which represent duration values), a small amount of
476 random variation was added to the location of each data point using the `jitter`
477 function.



478

479 Figure 7: Scatter plot of velocity and log-duration in frames for the closing phase of
480 Pause vs. Nasal segments/intervals in English and French

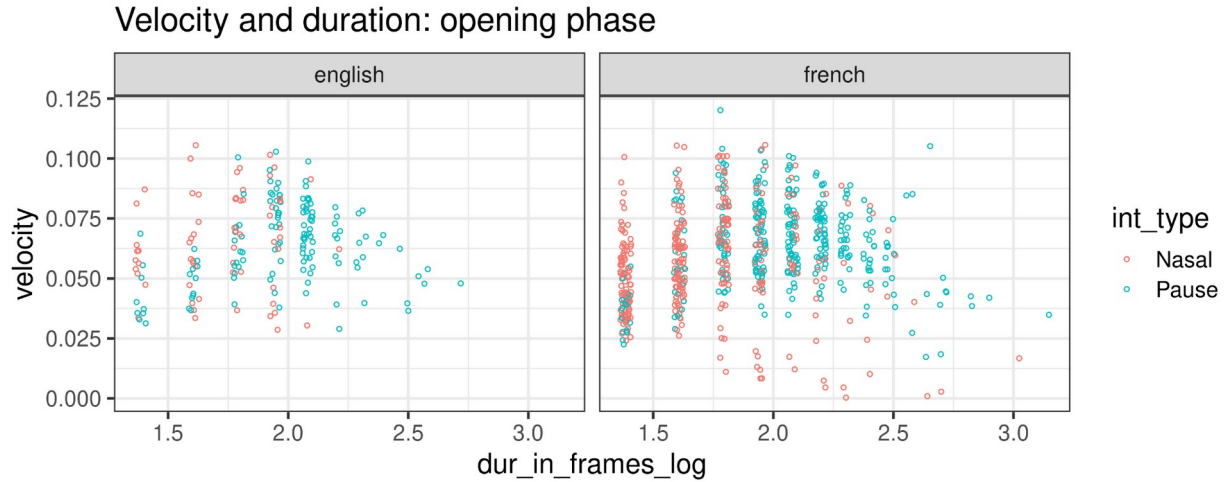


Figure 8: Scatter plot of velocity and log- duration in frames for the opening phase of Pause vs. Nasal segments/intervals in English and French

As Figures 7 and 8 show, a linear relationship is hard to discern between the log-duration and velocity; instead, there appears to be a slight indication of a parabolic relationship which is more pronounced for Pause intervals both for opening and closing phases. Velocity appears to be the highest (for both nasals and pauses) when the log of the duration in frames is near 2 (which represents a duration of 7 frames) and it gradually decreases as the duration gets longer. Contrarily, velocity is smaller when the log-duration gets shorter. The data points in the bottom-left corner (the smallest duration values) might be associated with shorter travel distance which could explain why these tokens have lower velocity. As an example, Figure 9 shows the data for French Pause intervals (during the closing phase) where the size and color depth of a dot reflects the token's travel distance value (i.e., a smaller and darker dot has a smaller travel distance). As the figure shows, most of the dots around the bottom-left corner (approximately below the hand-drawn dotted line) are smaller in size on average than the other dots; this indicates that the tokens having a shorter duration and lower velocity tend to have a smaller travel distance.

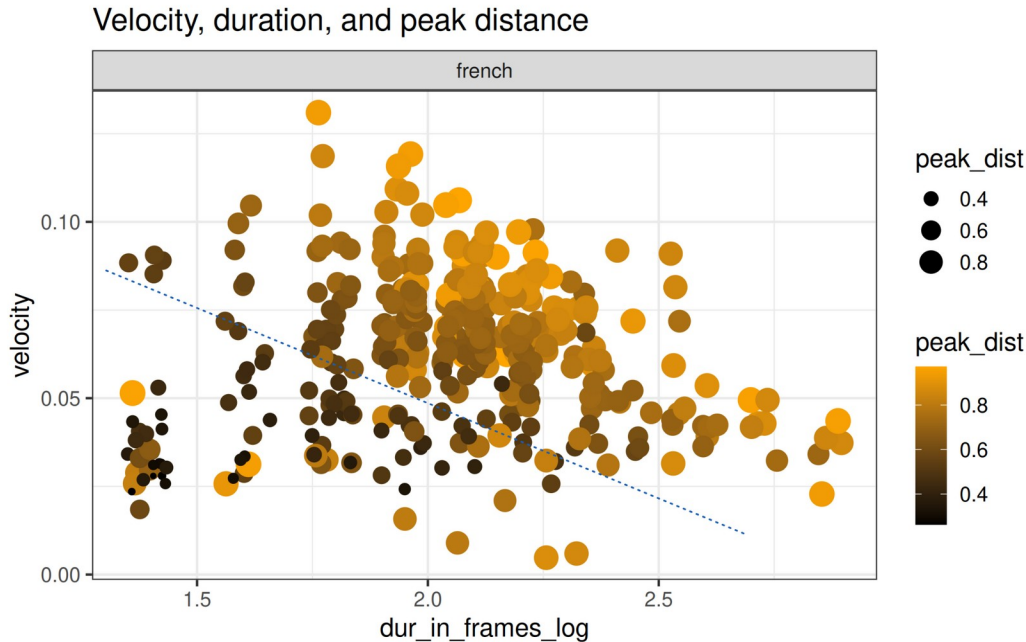


Figure 9: Velocity, duration, and travel distance in French Pause intervals (closing phase); the size of a data point reflects its travel distance value.

3.3 Linear mixed-effects models

We used R (R Core Team 2024) and *lmerTest* package (Kuznetsova, et al. 2017) to assess the statistical significance for the effects interval type, travel distance, duration, etc. on velum movement velocity using linear mixed-effects models (more details in Section 2. 2 above). Table 6 reports the relevant statistics for the predictor variables; the p-value for the level of significance (the α) was set at 0.05. The variables sex and phase (opening/closing) were dropped from the final model because the initial model confirmed their effects were not significant [phase (df = 1713, t 0.42, p = .672), sex (df = 13.35, t = 0.16, p = .879)].

Table 6: Coefficients from linear mixed-effects model of velar travel distance, duration (in log of frame count) of velar opening and closing movement in English and French for nasal and pause intervals.

predictor	estimate	std_err	df	t	p_value
(Intercept)	0.067	0.002	22.09	32.10	<.001
travel distance	0.034	0.001	1720.42	30.84	<.001
dur_in_frames_log	-0.018	0.001	1721.12	-	<.001
I(dur_in_frames_l	-0.014	0.001	1721.23	-	<.001

og ²)				11.36	
language-french	-0.001	0.003	21.25	-0.56	0.581
int_type-pause	-0.006	0.002	1722.72	-4.11	<.001
Language-french: int_type-pause	0.004	0.002	1713.91	2.31	0.021

517

518 As Table 6 reveals, both the numeric predictors (travel distance and duration in frames)
519 had statistically significant effects on the velocity variable. For travel distance, there was
520 a positive relationship such that the greater the distance the velum traveled, the greater
521 the velum movement velocity ($\beta = 0.034$, $df = 1720.42$, $t = 30.84$, $p < .001$). The
522 relationship was the opposite for duration; as the duration of an interval increased, the
523 velocity decreased ($\beta = -0.018$, $df = 1721.12$, $t = -16.93$, $p < .001$). Since these variables
524 were normalized using the same scale, the coefficients were comparable between
525 themselves, and a higher absolute value for travel distance (0.034) than duration
526 (0.018) indicated that travel distance had a greater impact than duration on velum
527 movement velocity. It is to be noted, however, that we also included a quadratic
528 polynomial term for the duration variable in the model to verify if a parabolic function
529 could add to our explanation of the data (this was inspired by the U-shaped patterns
530 observed in Figures 7 and 8. As the coefficient table shows, the quadratic term was
531 indeed statistically significant ($\beta = -0.014$, $df = 1721.23$, $t = -11.36$, $p < .001$). Thus, both
532 the linear and non-linear effects of duration on velum movement velocity were
533 significant; and the combined effect from the linear and non-linear functions of duration
534 ($\beta = 0.018 + 0.014 = 0.32$) was approximately equal to the effect of travel distance ($\beta =$
535 0.034).

536 More crucially, the model also confirms a significant effect of interval type (Nasal vs.
537 Pause) on velum movement velocity; velum moves slower during pause intervals than
538 during nasal segments ($\beta = -0.006$, $df = 1722.72$, $t = -4.11$, $p < .001$). The fixed main
539 effect of language was not statistically significant ($p = .581$); however, an interaction
540 term between language and interval type was statistically significant ($\beta = -0.004$, df
541 $= 1713.91$, $t = 2.31$, $p = .021$). Figure 10 shows the interaction effect; it can be seen that
542 the effect of language depends on the interval type (and vice-versa). The two languages
543 are very different within the nasal category; that is, the velum moves faster in English
544 nasals compared to French nasals. It can also be observed that French has a larger
545 difference in velocity between nasals and pauses compared to English.

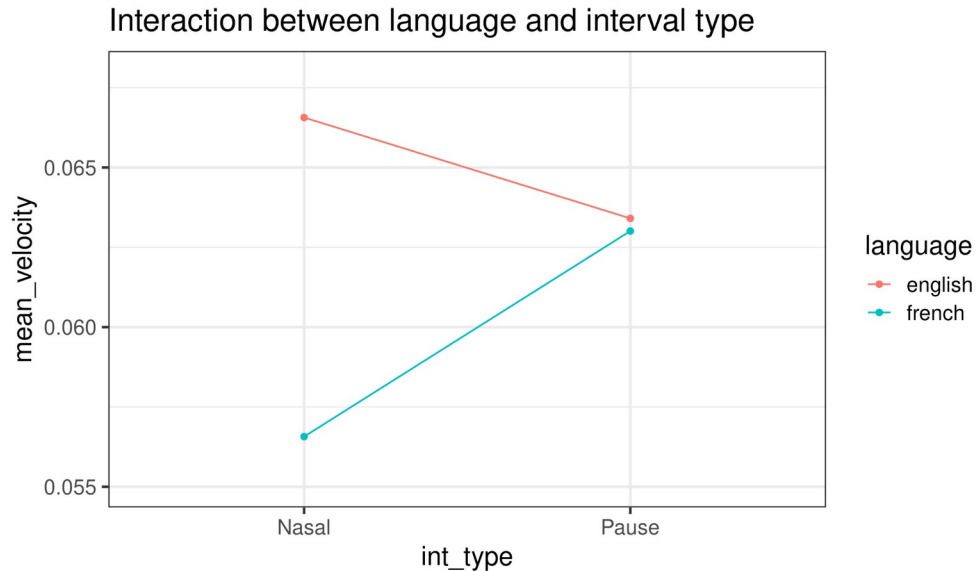


Figure 10: Effect of the interaction between language and interval type on velocity

4 Discussion

This study sought to investigate whether velum behaves differently in speech-related (Nasal segments) vs. speech-independent events (inter-speech Pauses). The exploratory investigations as well as the mixed-effects models reported above confirm that velum movement velocity does vary between inter-speech pauses and speech segments for specific languages. More specifically, the velum moved faster for an opening or closing of the velopharyngeal port (VPP) during a Pause interval compared to a speech (Nasal) interval in French. However, the effect could not be generalized to English (i.e., English pauses were not found to be different from English nasals). Therefore, it can be concluded that the velocity of velum movement in inter-speech pauses (vs. speech segments) is affected by the sound inventory of a language and its phonetic implementations.

Velum opening and closing gestures during inter-speech pauses were also confirmed to have longer duration compared to nasal segments, and this pattern was consistent for both French and English speakers. This could be an indication that the velum is less active during a pause interval due to reduced demand. The effect, however, is not straightforward since travel distance can interfere with duration, combined with the quadratic effect of duration, as shown in Figures 7, 8, 9, and Table 6.

The findings regarding the relationship between the velocity of the velum and the velum's travel distance and duration were interesting since both velum travel distance and duration were expected to affect velum movement velocity and the two factors can actually interact. Our results confirmed that both travel distance and duration had a

significant effect on velum movement velocity: the velum moved faster when the travel distance was greater; the velum moved slower when the segment was longer. This finding confirms our prediction that English nasals involve greater velum movement velocity than French nasals, which is consistent with the previous reports of English nasals having to involve a greater travel distance (VPO) than French nasals. If two segments have the same travel distance (i.e., VPO) of the velum, the one with a shorter duration will have faster velum movement; contrarily, if two segments are of the same duration, the one with a greater travel distance (greater peak VPO) will have faster velum movement. There is one caveat about this interaction between travel distance and duration though. The negative correlation between velocity and duration was noticeable only in the cases when the travel distance was big enough (approximately when the peak VPO was greater than 0.5, as can be approximated from Figure 9). The effect of duration was the opposite when the travel distance was small (indicated by the significance of the parabolic function in the linear model we presented above); that is, the velocity increased as the duration increased, which is counterintuitive. We conclude that our model can predict the velocity better for cases when the travel distance (VPO) is greater than 0.5; otherwise, we have less consistent and reliable predictions.

Alongside the general effects of inter-speech pauses on velum movement velocity and its movement duration, a number of other findings can be related to the results reported in previous studies. First, we could not confirm any difference between the opening and closing velocities for velum movement, as reported in Birkholz and Kleiner (2021) or the related measures of raising and lowering the velum reported by Kuehn (1976) and Benguerel et al. (1977a). Our beginning assumption was that the velum should move faster when closing since the closing phase should involve an articulatory overshoot to ensure a tight seal during oral segments, swallowing, etc. (while for opening, the lowering of the velum is assisted by gravity). The results, however, did not confirm a difference in velocity between the two phases.

Our results also revealed that English nasal speech segments involved faster velum movement than French nasals. This is consistent with our initial assumption that the velum would move faster in English nasals than in French nasals because English nasals have been reported to have a larger VPO than French nasals (Islam et al. 2020). This further confirms that velum behavior in speech can be affected by language differences.

5 Conclusion

This study investigated how velum movement differs between speech-related and speech-independent events in French and English, with a focus on movement velocity, travel distance, and duration. The findings show that velum speed and timing are

609 sensitive to both linguistic and biomechanical factors. Notably, French speakers
610 demonstrated faster velum movements during pauses compared to nasal segments, a
611 pattern not observed in English. Additionally, English nasals involved faster velum
612 movement than French, consistent with their larger VPO targets.

613 We also found that movement velocity is jointly influenced by travel distance and event
614 duration, with nonlinear interactions that complicate straightforward interpretation.
615 These results emphasize the importance of considering both linguistic context and
616 articulatory constraints when analyzing velum behavior.

617 Our study contributes to a better understanding of the dynamic nature of velopharyngeal
618 control and lays the groundwork for future research. Expanding this line of work to
619 include additional languages, more diverse speaker populations, and higher-resolution
620 imaging methods will be crucial for furthering our knowledge of speech motor control
621 and its clinical, modeling, and technological applications.

622 Despite its strengths, the study has some limitations. Our sample size, while larger than
623 that of many earlier studies, is still modest. We treated the sagittal VPO as a proxy for
624 velum displacement, but VPO may also be influenced by other structures such as the
625 superior pharyngeal constrictor or Passavant's ridge (Perry 2011). Finally, the findings
626 are limited to Quebecois French and Canadian English; broader cross-linguistic
627 validation remains necessary.

628

629 **Ethics statement**

630 This study utilized secondary data obtained from an existing and publicly available
631 database. No primary data were collected, and the researchers did not have access to
632 any personally identifiable information of the subjects.

633 **Author contributions**

634 Md Jahurul Islam contributed to the conceptualization of the research problem, data
635 preprocessing and analysis, and manuscript writing. Gillian de Boer contributed to the
636 literature review, critically evaluated data processing and results, and co-wrote the
637 manuscript. Bryan Gick provided overall project supervision, offering critical feedback
638 on the conceptual framework, data analysis, results, and manuscript editing.

639 **Conflict of interest statement**

640 The authors have no conflicts of interest to report.

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